

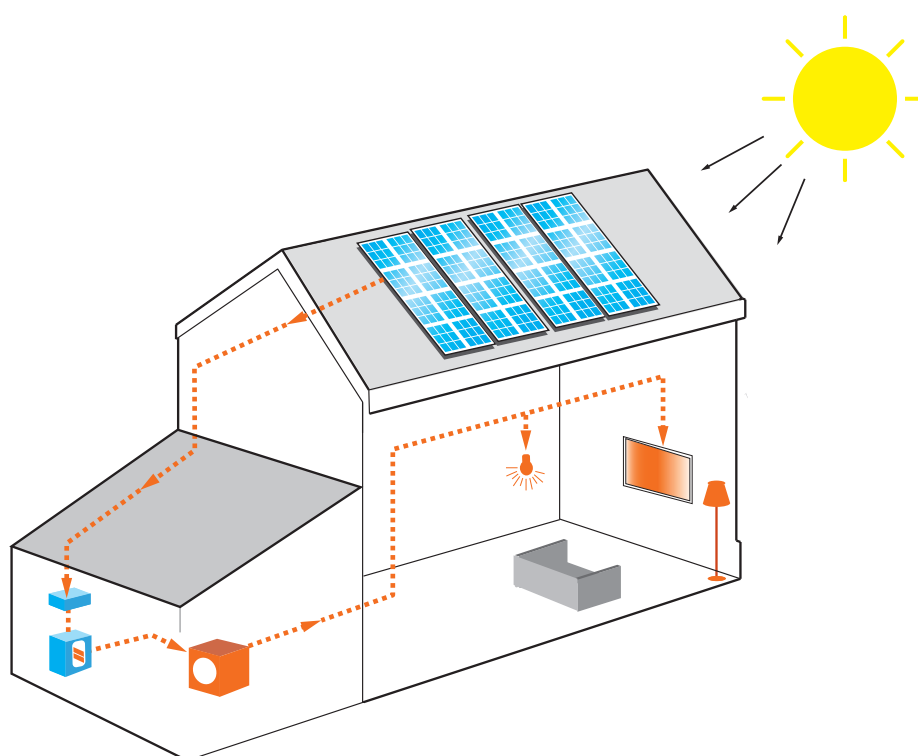
4. In the UK there are now a range of companies that will offer to install solar panels to homes.

Adverts like the one below are becoming quite common.

Sunny Sun Panels

You can save an average of £50 each month on your fuel bill!

When the Sun's radiation hits your solar panels it is converted into electrical energy. This energy travels to a device called an inverter, which converts it into usable household electricity. With over 1500 hours of sunshine per year what could be better!



Your own Sunny Sun panel system fully installed for £3800! What a price!
Add battery storage at only £2900 – you can use your solar electricity whenever you like.

Get your money back in 10 years!

Designed to work alongside your existing electricity supplier or as part of a new combined home solar system, battery storage will make it much easier to save on your electricity bills by utilising more of the electricity generated by your solar panels. This system will satisfy all your home electrical needs.

The excess power from your solar panels charges your battery during the day. The battery then supplies electricity to your home during the night and on cloudy days. This maximises your electricity savings and makes your house free from electricity from the National Grid.



- (a) In January 2018 a householder invested in the Sunny Sun panel system and battery storage.

- (i) Use the equation

$$\text{payback time} = \frac{\text{installation costs}}{\text{annual savings}}$$

and the information provided to determine whether the claim 'get your money back in 10 years' is true. [3]

- (ii) State **three** reasons why payback time might be more than expected. [3]

- (b) In 2018 the mean household electricity consumption for the UK was 4420 kWh per year.

An independent energy expert assessed the system and found they absorb a mean power of 14 000 W of sunlight and converted this to electrical power with an efficiency of 23%.

Sunny Sun claim that their panel system can satisfy the electrical needs of the average home in the UK.

Use this information together with information in the advert and the equation

$$\text{units produced (kWh)} = \text{power (kW)} \times \text{time (h)}$$

to determine if the Sunny Sun claim is valid. [4]



- (c) One of the largest solar power plants proposed in the UK is being developed in Kent. This solar power plant is capable of generating a maximum of 3.5×10^6 kWh per year. At present the UK generates 5.3×10^{10} kWh of electricity per year.

Use the information provided to:

- (i) Calculate how many homes this solar power plant could supply with electricity assuming that the mean household electricity consumption is 4420 kWh per year. [2]

number of homes =

- (ii) Calculate how many solar power plants of this size would be required to supply the UK electrical energy demand. [2]

number of plants =

